

VINCENT DO

Open to relocation · 779-207-6818 · vincentdo306@gmail.com · <https://linkedin.com/in/vincent-do-uiuc> · vmd306.com

EXPERIENCE

Boeing Company

Aug 2024 – Present

Software Engineer (Secret Security Clearance)

Berkeley, MO

- Shipped production C++ features delivering **6+ real-time navigation message streams** to display applications, backed by automated tests and Dockerized build workflows
- Designed the message protocol and internal C++ SDK adopted by **10+ engineers**, reducing duplicate integration code across downstream modules that rely on the team's messaging framework
- Own integration and release readiness for the shared framework used by **60+ dependent modules**, resolving timing-drift and binary-compatibility issues before release
- Led a Linux-based performance evaluation of a partner system using real-time flight data over TCP/UDP, sustaining **1.5× required throughput** across repeatable load tests
- Diagnosed and fixed a long-standing cross-system defect caused by a binary data-merge edge case within two weeks of joining the team, restoring correct downstream output
- Established the project's first automated test environment and built **20+ .NET flight-data simulations** incorporated into program-wide nightly test runs, providing repeatable end-to-end validation for new capabilities

Expedia Group

May 2023 – Aug 2023

Software Development Engineer Intern

Chicago, IL

- Shipped GraphQL schema changes across **5+ Kotlin microservices** to production, with unit tests covering new booking flows on iOS and Android
- Built backend support for localized search across **16 languages**, creating mock GraphQL responses to validate dynamic rendering with frontend and localization teams before release
- Shipped VIP Access badges, **10+ amenity filters**, and clearer error handling for mobile search; evaluated badge behavior through A/B testing

University of Wisconsin–Madison

May 2021 – Aug 2021

Biomedical Undergraduate Researcher

Madison, WI

- Analyzed **16,813 genes across 169 phenotypes** in Python and R, using recursive graph traversal, frequency thresholds, and chromosomal filtering to identify **25 high-priority Zellweger candidate genes**

PROJECTS

Airport Routing Service | C++20, REST API, Multithreading, Sockets, CMake/CTest, Docker [live demo](#) | [code](#)

- Built and deployed a C++20 HTTP/JSON airport-routing service over raw sockets, using BFS and Dijkstra for route search and precomputed centrality across **500 airports and 16,379 directed routes**
- Implemented a fixed worker pool and bounded request queue to manage concurrent traffic; local Release-build tests on a larger graph sustained **1,650+ requests/sec** with p99 latency below **120 ms** and no failures
- Modernized a legacy graph core from C++11 to C++20, fixing **heap-corruption and recursive-copy defects**; added **GitHub Actions CI** with CMake/CTest, memory sanitizers, API checks, and a Docker smoke test

Exploding Chickens | Node.js, MongoDB

[live demo](#)

- Built a real-time multiplayer backend in Node.js with asynchronous game logic and MongoDB-backed session state, supporting concurrent play across **1,600+ completed games**

Playable Chess AI | Python, PyTorch, CUDA, Flask/Gunicorn, Hugging Face

[live demo](#) | [code](#)

- Built a PyTorch/CUDA training pipeline that processed roughly **393M chess positions**, using mixed precision, cutting per-epoch training time **60×**, from 20+ hours on CPU to 20 minutes on GPU
- Built and deployed a Flask/Gunicorn chess service serving a **15.1M-parameter ResNet–LSTM** with alpha-beta search, hardened for concurrent load with readiness/liveness probes and a GitHub Pages frontend

TECHNICAL SKILLS

Languages: C++ (C++20), Python, Java, Kotlin, C#, SQL, R, TypeScript, JavaScript

Backend & Data: REST APIs, GraphQL, HTTP/1.1, TCP/UDP sockets, multithreading, thread pools, backpressure, Node.js, PostgreSQL, MongoDB, API and protocol design

Infrastructure: Docker, CMake/CTest, GitHub Actions, Jenkins, CI/CD, Linux, Fly.io, GCP, Git

Testing: Unit, integration, API, and load testing; Catch2; ASan/UBSan; health and readiness checks

EDUCATION

University of Illinois Urbana-Champaign

Aug 2020 – May 2024

B.S. Computer Science & Chemistry · GPA: 3.6

Champaign, IL